MSG No. 167.104 $R\bar{3}c1'$ [Type II, trigonal]

Table 1: Wyckoff site: `6a`, site symmetry: `32.1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,37,38,39,40,41,42]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10,11,12,43,44,45,46,47,48]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{7}{12}\right]$	`[13,14,15,16,17,18,49,50,51,52,53,54]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{12}\right]$	`[19,20,21,22,23,24,55,56,57,58,59,60]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{11}{12}\right]$	`[25,26,27,28,29,30,61,62,63,64,65,66]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{5}{12}\right]$	`[31,32,33,34,35,36,67,68,69,70,71,72]`

Table 2: Wyckoff site: `6b`, site symmetry: `-3..1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,7,8,9,37,38,39,43,44,45]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,10,11,12,40,41,42,46,47,48]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$	`[13,14,15,19,20,21,49,50,51,55,56,57]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[16,17,18,22,23,24,52,53,54,58,59,60]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$	`[25,26,27,31,32,33,61,62,63,67,68,69]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[28,29,30,34,35,36,64,65,66,70,71,72]`

Table 3: Wyckoff site: `12c`, site symmetry: `3..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,37,38,39]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,6,40,41,42]`
3	$[0,\ 0,\ -z]$	`[7,8,9,43,44,45]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12,46,47,48]`
5	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$	`[13,14,15,49,50,51]`
6	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}-z\right]$	`[16,17,18,52,53,54]`
7	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}-z\right]$	`[19,20,21,55,56,57]`
8	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{5}{6}\right]$	`[22,23,24,58,59,60]`
9	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$	`[25,26,27,61,62,63]`
10	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}-z\right]$	`[28,29,30,64,65,66]`
11	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}-z\right]$	`[31,32,33,67,68,69]`
12	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{6}\right]$	`[34,35,36,70,71,72]`

Table 4: Wyckoff site: `18d`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,7,37,43]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,8,38,44]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,9,39,45]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,10,40,46]`
5	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,11,41,47]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,42,48]
7	$\left[\frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right]$	[13,19,49,55]
8	$\left[\frac{2}{3}, \frac{5}{6}, \frac{1}{3}\right]$	[14,20,50,56]
9	$\left[\frac{1}{6}, \frac{5}{6}, \frac{1}{3}\right]$	[15,21,51,57]
10	$\left[\frac{1}{6}, \frac{1}{3}, \frac{5}{6}\right]$	[16,22,52,58]
11	$\left[\frac{2}{3}, \frac{5}{6}, \frac{5}{6}\right]$	[17,23,53,59]
12	$\left[\frac{1}{6}, \frac{5}{6}, \frac{5}{6}\right]$	[18,24,54,60]
13	$\left[\frac{5}{6}, \frac{2}{3}, \frac{2}{3}\right]$	[25,31,61,67]
14	$\left[\frac{1}{3}, \frac{1}{6}, \frac{2}{3}\right]$	[26,32,62,68]
15	$\left[\frac{5}{6}, \frac{1}{6}, \frac{2}{3}\right]$	[27,33,63,69]
16	$\left[\frac{5}{6}, \frac{2}{3}, \frac{1}{6}\right]$	[28,34,64,70]
17	$\left[\frac{1}{3}, \frac{1}{6}, \frac{1}{6}\right]$	[29,35,65,71]
18	$\left[\frac{5}{6}, \frac{1}{6}, \frac{1}{6}\right]$	[30,36,66,72]

Table 5: Wyckoff site: 18e, site symmetry: .2.1'

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	[1,4,37,40]
2	$\left[0, x, \frac{1}{4}\right]$	[2,5,38,41]
3	$\left[-x, -x, \frac{1}{4}\right]$	[3,6,39,42]
4	$\left[-x, 0, \frac{3}{4}\right]$	[7,10,43,46]
5	$\left[0, -x, \frac{3}{4}\right]$	[8,11,44,47]
6	$\left[x, x, \frac{3}{4}\right]$	[9,12,45,48]
7	$\left[x+\frac{2}{3}, \frac{1}{3}, \frac{7}{12}\right]$	[13,16,49,52]
8	$\left[\frac{2}{3}, x+\frac{1}{3}, \frac{7}{12}\right]$	[14,17,50,53]
9	$\left[\frac{2}{3}-x, \frac{1}{3}-x, \frac{7}{12}\right]$	[15,18,51,54]
10	$\left[\frac{2}{3}-x, \frac{1}{3}, \frac{1}{12}\right]$	[19,22,55,58]
11	$\left[\frac{2}{3}, \frac{1}{3}-x, \frac{1}{12}\right]$	[20,23,56,59]
12	$\left[x+\frac{2}{3}, x+\frac{1}{3}, \frac{1}{12}\right]$	[21,24,57,60]
13	$\left[x+\frac{1}{3}, \frac{2}{3}, \frac{11}{12}\right]$	[25,28,61,64]
14	$\left[\frac{1}{3}, x+\frac{2}{3}, \frac{11}{12}\right]$	[26,29,62,65]
15	$\left[\frac{1}{3}-x, \frac{2}{3}-x, \frac{11}{12}\right]$	[27,30,63,66]
16	$\left[\frac{1}{3}-x, \frac{2}{3}, \frac{5}{12}\right]$	[31,34,67,70]
17	$\left[\frac{1}{3}, \frac{2}{3}-x, \frac{5}{12}\right]$	[32,35,68,71]
18	$\left[x+\frac{1}{3}, x+\frac{2}{3}, \frac{5}{12}\right]$	[33,36,69,72]

Table 6: Wyckoff site: 36f, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,37]
2	$[-y, x-y, z]$	[2,38]
3	$[-x+y, -x, z]$	[3,39]
4	$\left[x-y, -y, \frac{1}{2}-z\right]$	[4,40]
5	$\left[y, x, \frac{1}{2}-z\right]$	[5,41]

continued ...

Table 6

No.	position	mapping
6	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[6,42]
7	$[-x,\, -y,\, -z]$	[7,43]
8	$[y,\, -x+y,\, -z]$	[8,44]
9	$[x-y,\, x,\, -z]$	[9,45]
10	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[10,46]
11	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[11,47]
12	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[12,48]
13	$\left[x+\frac{2}{3},\, y+\frac{1}{3},\, z+\frac{1}{3}\right]$	[13,49]
14	$\left[\frac{2}{3}-y,\, x-y+\frac{1}{3},\, z+\frac{1}{3}\right]$	[14,50]
15	$\left[-x+y+\frac{2}{3},\, \frac{1}{3}-x,\, z+\frac{1}{3}\right]$	[15,51]
16	$\left[x-y+\frac{2}{3},\, \frac{1}{3}-y,\, \frac{5}{6}-z\right]$	[16,52]
17	$\left[y+\frac{2}{3},\, x+\frac{1}{3},\, \frac{5}{6}-z\right]$	[17,53]
18	$\left[\frac{2}{3}-x,\, -x+y+\frac{1}{3},\, \frac{5}{6}-z\right]$	[18,54]
19	$\left[\frac{2}{3}-x,\, \frac{1}{3}-y,\, \frac{1}{3}-z\right]$	[19,55]
20	$\left[y+\frac{2}{3},\, -x+y+\frac{1}{3},\, \frac{1}{3}-z\right]$	[20,56]
21	$\left[x-y+\frac{2}{3},\, x+\frac{1}{3},\, \frac{1}{3}-z\right]$	[21,57]
22	$\left[-x+y+\frac{2}{3},\, y+\frac{1}{3},\, z+\frac{5}{6}\right]$	[22,58]
23	$\left[\frac{2}{3}-y,\, \frac{1}{3}-x,\, z+\frac{5}{6}\right]$	[23,59]
24	$\left[x+\frac{2}{3},\, x-y+\frac{1}{3},\, z+\frac{5}{6}\right]$	[24,60]
25	$\left[x+\frac{1}{3},\, y+\frac{2}{3},\, z+\frac{2}{3}\right]$	[25,61]
26	$\left[\frac{1}{3}-y,\, x-y+\frac{2}{3},\, z+\frac{2}{3}\right]$	[26,62]
27	$\left[-x+y+\frac{1}{3},\, \frac{2}{3}-x,\, z+\frac{2}{3}\right]$	[27,63]
28	$\left[x-y+\frac{1}{3},\, \frac{2}{3}-y,\, \frac{1}{6}-z\right]$	[28,64]
29	$\left[y+\frac{1}{3},\, x+\frac{2}{3},\, \frac{1}{6}-z\right]$	[29,65]
30	$\left[\frac{1}{3}-x,\, -x+y+\frac{2}{3},\, \frac{1}{6}-z\right]$	[30,66]
31	$\left[\frac{1}{3}-x,\, \frac{2}{3}-y,\, \frac{2}{3}-z\right]$	[31,67]
32	$\left[y+\frac{1}{3},\, -x+y+\frac{2}{3},\, \frac{2}{3}-z\right]$	[32,68]
33	$\left[x-y+\frac{1}{3},\, x+\frac{2}{3},\, \frac{2}{3}-z\right]$	[33,69]
34	$\left[-x+y+\frac{1}{3},\, y+\frac{2}{3},\, z+\frac{1}{6}\right]$	[34,70]
35	$\left[\frac{1}{3}-y,\, \frac{2}{3}-x,\, z+\frac{1}{6}\right]$	[35,71]
36	$\left[x+\frac{1}{3},\, x-y+\frac{2}{3},\, z+\frac{1}{6}\right]$	[36,72]